

JEONGWOO SHIN

Updated: July 28, 2026

Email: swsyss@snu.ac.kr Website: swsyss.github.io Phone: +82 10-9555-9713

RESEARCH INTEREST

Generative Modeling, Diffusion & Flow Matching, Reward-Finetuning of Diffusion Models, Stochastic Optimal Control, Diffusion/Flow Model Distillation, Long Video Generation, 3D-Consistent Video Generation.

EDUCATION

Graduate School of Data Science, Seoul National University (SNU)

Ph.D. Data Science

2024.03 – Current

- Advisor: Joonseok Lee
- Research: Generative Modeling, Diffusion & Flow Matching
- GPA: 3.70 / 4.30

Graduate School of Data Science, Seoul National University (SNU)

M.S. Data Science

2022.03 – 2024.02

- Thesis: Self-Guided Masked Autoencoder
- Advisor: Joonseok Lee
- GPA: 3.98 / 4.30

Seoul National University (SNU)

B.S. Landscape Architecture and Rural System Engineering

2016.03 – 2022.02

- Thesis: Adsorptive Removal of Methylene Blue from Aqueous Solution Using Carbon Nanospheres
- GPA: 3.59 / 4.30

PUBLICATIONS

* Equal contribution. † Corresponding author.

[C4] Efficient Generative Modeling beyond Memoryless Diffusion via Adjoint Schrödinger Bridge Matching

[Jeongwoo Shin](#), Jinhwan Sul, Joonseok Lee[†], Jaewoong Choi[†], Jaemoo Choi[†]

International Conference on Machine Learning (ICML) 2026

- Solved Schrödinger Bridge problem (dynamic optimal transport) via optimal control perspective and reached high stability and performance in image generation.

[C3] Equivariant Latent Alignment via Flow Matching under Group Symmetries

Sunghyun Kim*, Jaehoon Hahm*, [Jeongwoo Shin](#), Joonseok Lee[†]

International Conference on Machine Learning (ICML) 2026

- Utilized Flow model as distribution bridging method for correction in equivariant latent space.

[C2] Latent Diffusion Models with Masked AutoEncoders

Junho Lee*, [Jeongwoo Shin](#)*, Hyungwook Choi, Joonseok Lee[†]

International Conference on Computer Vision (ICCV) 2025

- Suggested Masked Autoencoder-based tokenizer for better performance in latent diffusion models.

[C1] Self-Guided Masked Autoencoder

[Jeongwoo Shin](#), Inseo Lee, Junho Lee, Joonseok Lee[†]

Conference on Neural Information Processing Systems (NeurIPS) 2024

- Figured out that Masked Autoencoder learns feature separation from early training stage, and proposed self-derived informative masking strategy for faster convergence.

[P4] Efficient Adjoint Matching for Fine-tuning Diffusion Models

Jeongwoo Shin*, Dongsoo Shin*, Yuchen Zhu, Wei Guo, Yongxin Chen, Joonseok Lee†, Jaewoong Choi†, Jaemoo Choi†

Under Review at NeurIPS 2026

- Reformulated the base dynamic of the pretrained text-to-image Diffusion model for efficient adjoint-based reward fine-tuning.

[P3] Diverse Motion Customization via Control-based Dynamic Optimization

Youngyoon Choi*, Kihyun Kim*, Jeongwoo Shin†, Joonseok Lee†

Under Review at NeurIPS 2026

- Converted the motion customization task to dynamic-optimization regime, and suggested control-based framework to prevent content leakage problem in this task.

[P2] No More Hit-and-Hope: Optimal Initial Noise Forging via Noise Shuffling

Suho Ryu, Jeongwoo Shin, Joonseok Lee†

Under Review at NeurIPS 2026

- Introduced the test-time prior-modification method for better text-aligned image generation.

[P1] Optimal Crystal Flow for Fast and Reliable Crystal Generation

Seunghoon Yi, Youngwoo Cho, Jaehoon Hahm, Jeongwoo Shin, Soo Kyung Kim†, Jaegul Choo†, Joonseok Lee†, Hongkee Yoon†

Under Review at NeurIPS 2026

- Tailored the mini-batch optimal transport-based Flow to Crystal domain for faster and more accurate Crystal generation.

AWARDS &
SCHOLARSHIPS

Research Fellowship for Ph.D. Candidates

2025.09 – 2026.08

Principal Investigator (PI) National Research Foundation of Korea (NRF)

- Project: *Optimal Latent Space for Effective Image Generation*

- Competitive individual research grant for Ph.D. students, funded by the Ministry of Education.

RESEARCH
EXPERIENCE

Data Refinement using Multi-Modal Foundation Models

2023.11 – 2024.09

Samsung Electronics Co., Ltd.

- Collaborative research project with Samsung Electronics.

- Worked on data refinement with large multi-modal foundation models for personalized search and generation.

TEACHING
EXPERIENCE

TA, SNU

M3239.005300: Machine Learning and Deep Learning I

2023 Spring

LANGUAGES

Programming: Python, C, C++

Deep Learning: PyTorch, TensorFlow, NumPy

Natural Languages: English (Fluent), Korean (Native)

REFERENCES

Joonseok Lee
M.S. & Ph.D. Advisor (SNU)

2022.03 - Current
joonseok@snu.ac.kr